Self-Transcending Generators:
Fixed Causal Laws Without Final Explanatory Closure

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Abstract

A hidden creed governs much contemporary thought: if a phenomenon is generated by fixed fundamental laws, then that phenomenon must admit *final explanatory closure*—a fixed explanatory standpoint from which everything important about it can be said. The formal theory developed here shows that this creed is false in a sharply delimited but nontrivial sense. There are finitely specified lawful generators whose entire phase futures lie in a single causal trace, yet *fixed admissible explanatory closure* for the full tower is impossible within the relevant class of reducers.

What makes this result nontrivial is not any one familiar fact taken in isolation. It is not merely that complexity grows, that prediction can be hard, that vocabularies can change, that science revises its theories, or that formal systems can outrun particular proof resources. The proved conjunction is stronger: one fixed lawful generator; an infinite sequence of generated phases; conservative yet irreducible explanatory succession; systematic failure of every fixed reducer of the relevant class; and, at the crown, upward explanatory necessity, where later generated regimes become required for structural truths about the generator itself.

The theorem family is not a restatement of Gödelian incompleteness or Tarskian undefinability [3, 6]: those limit provability and truth-definition in general logical settings; here the tension is *final explanatory closure* for towers of phases and regimes under a class of reducers. It is not computational irreducibility in the usual sense [8]—the scarcity of shortcuts to *compute* later states—because the issue here is not simulation cost but whether any *final explanatory stance* in the class is possible even when the trace is lawfully generated (and in principle simulatable). It does not reduce to Kuhnian historical paradigm change [4]: that framework is philosophical and narrative rather than a single formal statement; the "paradigm shift" used in this theory is a conservative yet irreducible regime transition set up to be proved, not a sociological episode. Generic emergence rhetoric and vague "science updates its vocabulary" slogans are likewise beside the quantified conjunction. The positive target is *explanatory anti-closure* under fixed law: exact causal generation does not force final explanatory closure, and explanatory order need not bottom out where causal generation bottoms out.

Philosophically, this yields a third option between naïve reductionism and anti-naturalist mystification: lawful self-transcending structure. Everything in the tower remains generated from below under fixed law, yet explanation need not close from below. The paper does not claim a universal diagonal over all raw interfaces, a universal paradigm tower for every regime family, backward causation, or that every deterministic system exhibits the phenomenon. It claims, instead, the existence, structure, and transferred stability of lawful self-transcending generators under the exact hypotheses stated in the cited formal results (Sections 3 and 18; Tables 1–3).

Formal proof in Lean 4. The claims anchored in `NoveltyTheory/` are machine-checked theorems, not prose: the project library currently contains **422** `theorem`/`lemma` declarations by the same automated line parse used in the repository audit (scope: `NoveltyTheory/` sources, not Mathlib), with **zero** `sorry` and **no** `axiom` declarations in those sources.

Keywords: lawful generation, explanatory closure, explanatory anti-closure, paradigm shift, proof-theoretic ascent, upward explanatory necessity, Lean formalization, admissible reducers, self-transcending generators.
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Contents

1 Introduction

> If a phenomenon is generated by fixed fundamental laws, then in principle it admits *final explanatory closure*: a fixed explanatory standpoint from which everything important about that phenomenon can be said.

That creed is one of the deepest unexamined assumptions in modern thought. The theory developed here overturns it by exhibiting a sharp mathematical profile and proving its stability under transfer.

A foreseeable first reaction is that none of this should surprise anyone: of course descriptions lag, of course theories mature, of course novelty accumulates. Section 4 and Section 5 address that reaction directly. The point is not to deny everyday experience, but to show that the *informal* recognition of open-ended inquiry is a different thing from the *formal joint profile* whose clauses are listed in machine-checked form: universal failure of *fixed* reducers of the relevant class on an infinite tower while a *single* law still generates every phase, conservative irreducible regime change, upward dependence for generator-structural truths at the crown, and transport under explicit hypotheses where proved. Without that conjunction pinned, one cannot license either the hidden creed *or* an equally sweeping denial of it.

The thesis is not that laws fail, that causation fails, or that later phases are ungenerated. It is sharper: a system can be causally complete without admitting final explanatory closure. A fixed law can produce every phase that appears and still defeat every would-be candidate for fixed admissible explanatory closure drawn from a nontrivial admissible class.

Formal backbone. The argument is paired with a machine-checked Lean 4 development whose cited declarations fix the exact formal scope of the claims proved here.

> **Scope and headlines.** *Honest scope:* not every deterministic system realizes full crown packaging; where predicates are absent, the theorems are silent; naive universal diagonals over raw interfaces are obstructed, not proved (Section 18). *Three headlines:* **(1)** no fixed final explanatory stance for the whole tower from a single closure candidate in the class (Section 9, Section 10); **(2)** generalized crown packaging—ascent, transfer, terminality defeat, alignment, stability (Section 11); **(3)** crown inversion—later regimes can be necessary for generator-structural truth (Section 12).

The results ascend through the following summits (expository order):

Summit I. **Triviality guard and schema.** What would be cheap, and what conjunction is actually targeted (Section 5, Section 6).

Summit II. **Object layer.** Self-transcending generators and conservative paradigm shift as formal objects (Section 7, Section 8).

Summit III. **Fixed generation without final explanatory closure.** Every phase on the trace, yet every fixed reducer in the class fails somewhere (Section 9).

Summit IV. **No final explanatory stance.** A single internal standpoint cannot close the generated future in the relevant class (Section 10).

Summit V. **Generalized final crown.** Class transfer, six crown clauses, naturality-aware transport where recorded (Section 11).

Summit VI. **Upward explanatory necessity.** Later regimes indispensable for generator-structural truths (Section 12).

Summit VII. **Orders, retro, contrasts, philosophy.** Explanatory vs. causal order, retroactive revelation, non-equivalences, and the third option (Section 13–Section 17).

Summit VIII. **Folklore vs. joint theorem shape.** Why commonplace open-endedness is not the same object as the certified configuration (Section 4, Section 5).

What would be unsurprising is a world in which outputs merely outstrip a fixed vocabulary, observers run out of time, or simulation remains expensive. The proved models go further: a single finitely specified generator can causally produce an infinite tower of phases and explanatory regimes such that *fixed admissible explanatory closure* fails for the full tower, and later generated regimes can become necessary for structural truths about the generator itself.

Section 3 records the main proved configuration in one place (theorem spine, six headline claims, boundaries). Expository order follows Summits I–VIII above and the table of contents. Section 2 is the entry point for mapping informal worries to formal quantifiers; proof sketches and certificate chains begin in Section A.

2 Premises, disagreement coordinates, and trust boundary

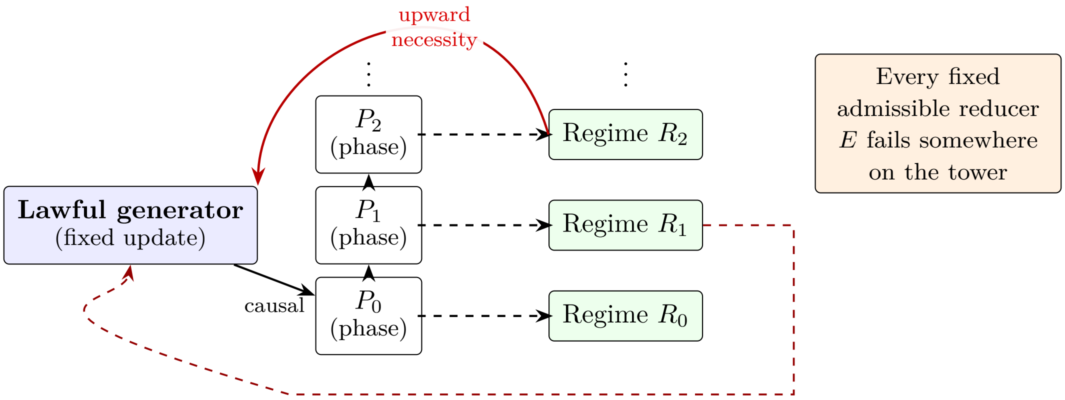

Figure 1: Conceptual spine: a fixed law drives phases upward in causal order; explanatory regimes attach to phases; every fixed admissible reducer fails somewhere on the tower; upward explanatory necessity relates later regimes to structural truths of the source. Broad transport across aligned natural carriers and organization-scale obstructions (as in Section 11 and Table 2) show the phenomenon is not encoding-local. Explanatory dependence need not mirror causal order.

Where disagreement can (and cannot) live

Formal spine. The generalized final crown package (six coordinated clauses), lower summits, obstructions, and naturality-aware transport are proved in `NoveltyTheory/` as cited; for a full machine-checked audit trail see Table 1–Table 3 and Section A.

Premise coordinates (for external readers).

- **Global metaphysics vs. formal quantifiers.** Theorems refute *fixed admissible explanatory closure relative to the stated class* in the constructed models and under the predicates named (including natural-class hypotheses where used). A blanket claim that "no final vocabulary is possible for all phenomena everywhere" is *not* what is discharged. *Logical effect:* Map polemics to Table 1, Table 2, and Section 18.

- **Causal completeness vs. final explanatory closure.** The paper denies that lawful generation forces final explanatory closure in the relevant sense; it does *not* deny forward causation or that phases lie in the generator's trace. *Logical effect:* Upward necessity (Section 12) concerns proof- and regime-relative access, not backward causation (Section 13).

- **Coding trick vs. class transfer.** A narrow "only one model" objection is met, under explicit hypotheses, by class transfer and the naturality-aware block. *Logical effect:* Compare the objection's quantifiers to Section 11 and Table 2.

- **Universal inevitability vs. cited predicates.** "Every lawful generator outruns every numeric bound" and "every weak paradigm step packages a full shift" are *false* under unrestricted quantifiers; what is proved are structured witnesses and, where relevant, categorical equivalences and obstructions at the stated layer (Section 18, Section A). *Logical effect:* Map sweeping glosses to Table 3 and the obstruction lemmas named there.

3 What is proved: spine, headlines, and boundaries

This section replaces an item-by-item abstract inventory with a reader-facing map; the machine-checked declarations are cited throughout and tabulated in Table 1–Table 3.

Four thematic peaks. A coordinated, machine-checked family organizes the spine as follows: (i) *Causal completeness without final explanatory closure*—every phase is generated forward, while each fixed reducer in the class fails somewhere; (ii) *Infinite conservative paradigm succession*—later regimes can preserve what earlier ones secured and still force irreducibly new structure; (iii) *Upward explanatory necessity* (the crown)—later regimes can be indispensable for structural truths about the generator itself; (iv) *Generalized transfer*—the pattern is not a coding accident of one model: disciplined transport across a nontrivial class of aligned systems, with naturality-aware strengthening where the predicates apply (Section 11, Table 2). A further layer records *sharp boundaries*: packaged paradigm succession aligns with a weak step *and* history conservativity at the formal packaging layer; on numeric observation type, unbounded observable recurrence lines up against uniform trace bounds in the recorded dichotomies; and unbounded recurrence is *not* a consequence of lawful generation alone (explicit counterexamples exist in the library). None of the positive peaks require indeterminism, backward causation, or blanket universals over arbitrary interfaces; quantifiers in the statements mark exactly what is—and is not—asserted.

Six central claims

1. Fixed law without a final explanatory stance. A single finitely specified generator can produce an infinite family of phases while no fixed explanatory regime drawn from the class uniformly captures them all.

2. Infinite conservative paradigm towers. Later regimes can preserve prior phase adequacy and still force genuinely new explanatory structure.

3. Upward explanatory necessity. Later generated regimes can become necessary to express or prove structural truths about the generator's own long-run behavior.

4. Lawful self-transcendence with generalized transfer. The architecture is neither anti-determinist nor merely rhetorical "emergence"; the flagship pattern survives transport beyond a single distinguished witness model.

5. Generalized transport under explicit natural hypotheses. Beyond the witness-level crown, the same strict-gap architecture extends across aligned carriers in the natural class and supports bounded-organization obstruction and anti-closure results under the stated naturality predicates.

6. Sharp boundaries at the formal layer. Conservativity in history is not redundant in the packaged paradigm-shift definition; numeric unbounded recurrence is a genuine load-bearing hypothesis, equivalent at that layer to non-uniform boundedness of the trace; and no blanket "universal escalation" of recurrence across *all* lawful numeric generators is claimed or holds (Section 18, Table 3).

4 Why this can sound obvious—and is not

Separating folklore from the proved configuration

What people already agree on (informally). Science revises its vocabulary; models are not prophets; interesting worlds overflow any fixed report at a fixed time; computational descriptions can be expensive; "there is always another question." None of these truisms, taken alone, is the target.

What is *not* already encoded in those truisms. The paper's mathematics concerns a *single* lawful generator whose trace includes every phase, paired with a *quantified* debacle for *every* explanatory reducer in a disciplined class, *together with* conservative non-trivial regime change and (under the crown hypotheses) upward necessity for truths about the generator itself. That *joint* possibility is stronger than "people innovate language" or "prediction is hard"; it is a structural statement about *explanatory anti-closure under exact generation*, not merely about human effort or historical contingency.

Why "we knew it all along" is an equivocation. It conflates (a) the sociological fact that institutions update theories with (b) the logical claim that fixed law + forward generation *entails* a final explanatory stance for the whole future from the relevant class, or with (c) the denial that any such stance is forced. (a) is beside the point; (b) is what the hidden creed needs; (c) is what the formal theorems refute in the constructed scopes. The mathematics is there to stop the slide between (a) and (c) without pretending to have proved a universal metaphysics.

What hard currency looks like here. Depth-indexed derivability gates that genuinely strengthen when depth increases; diagonal facts that are true on the canonical generator but unprovable at matching depth; twin carriers with aligned admissibility but trace mismatch; organization-height obstructions stated against explicit bounded-organization predicates; numeric equivalences that identify exactly when a "future defeat" hypothesis holds. These are not vibes: they are checks one can trace in code (Section A).

The point, then, is not that human inquiry is open-ended in the ordinary sense. The point is that a formally specified class of lawful generators can exhibit a much stronger joint profile: one source, total generation of the relevant future, conservative yet irreducible explanatory succession, systematic explanatory anti-closure—failure of fixed admissible explanatory closure—in the relevant class, and later-regime necessity for truths about the source itself. That conjunction is not common sense. It is what the formal theory isolates.

5 What would be trivial, and why this is not that

Readers may still worry that the whole undertaking collapses into a fancy restatement of humility. The next lists separate *cheap* phenomena from the *non-cheap* conjunction the paper actually establishes.

The theory would be uninteresting if it merely proved one of the following:

- a fixed language eventually fails on sufficiently rich outputs;
- complexity can outrun a bounded observer;
- prediction can become hard or impossible;
- a process can have infinitely many outputs;
- one can always invent richer descriptive vocabulary after the fact.

Each of these is real, but none is enough. None produces the kind of inversion the crown construction targets.

The actual target is a much stronger conjunction: a finitely specified generator G and a family of phases and regimes (P_n, R_n) such that

1. every P_n is generated by G;
2. each R_n adequately explains P_n in the formal sense of the theory;
3. each successor transition is conservative over prior history and irreducible to the prior regime;
4. every reducer in the admissible class fails to secure fixed admissible explanatory closure somewhere on the tower;
5. later regimes can become necessary for structural truths about the generator itself.

It is not a theorem of mere output richness, brute infinitude, or observational unpredictability alone. It is a theorem of *non-final explanation under fixed law.*

This is the sense in which the framework is a theory of lawful self-transcendence. The center of gravity is not "more happens than finite observers compress easily." It is: "fixed lawful generation does not guarantee a final explanatory stance." A single fixed generator can causally produce an infinite tower of phases and regimes such that none are uniformly collapsible into a single final explanatory stance of the relevant kind, and later regimes can become necessary for truths about the generator itself.

6 Causal generation versus explanatory reducibility

The core distinction is between two relations:

- **Causal generation:** the system produces a phase, state, or phenomenon as part of its lawful trace.
- **Explanatory reducibility:** a fixed admissible regime suffices to preserve the relevant structural distinctions, adequacy conditions, and invariant content.

These are not the same relation—and treating them as identical is exactly how the hidden creed smuggles in final closure.

To say that a phase is generated is a constructive or causal claim. To say that it is reducible to a fixed admissible regime is stronger: it requires structural adequacy without loss of the certified distinctions the theory cares about.

Definition 6.1 (Informal self-transcendence schema)**.** A generator G is self-transcending, in the core sense used here, if there exists a sequence $(P_n, R_n)_{n\in\mathbb{N}}$ such that:

$$\forall n,\ \mathsf{GeneratedBy}(G, P_n), \quad \forall n,\ \mathsf{Adeq}(R_n, P_n), \quad \forall n,\ \mathsf{ParadigmShift}(R_n, R_{n+1}; P_{\leq n}),$$

and

$$\forall E \in \mathsf{Adm},\ \exists n,\ \neg\mathsf{Expl}(E, G, P_n, R_n).$$

The force of this definition is easy to miss. The theorem is not merely that later phases need later words. It is that *every fixed admissible explanatory reducer fails somewhere.* This is the structural anti-closure statement.

Causal sufficiency does not imply final explanatory closure.

That sentence is the lens through which everything below should be read.

7 Self-transcending generators: a new species of object

The self-transcending generator is not a rhetorical flourish; it is the field-defining object of the theory. If such objects exist cleanly under honest admissibility hypotheses, the program is worth doing—not as cataloguing a formal artifact, but because they carve out a new structural possibility in the conceptual space of law, explanation, and reduction.

Why this is not "a complicated dynamical system." A generator with intricate outputs, a hard-to-predict future, or a long run of surprising behavior may still admit a fixed explanatory regime that is final relative to the admissible class. The self-transcending generator is characterized by the *failure* of that finitude: the causal trace is fully owned by one fixed law, while final explanatory closure is systematically defeated for every admissible reducer at some height on the tower. Complexity alone does not imply that joint profile.

Why the object is field-defining. Much of science and philosophy silently assumes that tightening law and tightening explanation march together. The self-transcending generator is the minimal schematic shape of their divergence: one fixed engine; inexhaustible stratification of adequate regimes; conservative, irreducible transitions; universal failure of any final explanatory stance to yield fixed admissible explanatory closure; and (at the crown) upward necessity for truths about the source itself. Theorems about this shape are not marginal remarks about pathological examples; they are theorems about what *lawful explanation* can and cannot promise.

Why existence is philosophically radical. If even one such generator fits the formal package, the hidden creed cannot be treated as analytic. It becomes an empirical-conceptual hostage: perhaps nature never realizes the package, but *in principle* fixed law without final explanation is coherent, structured, and provable. That shifts the default: final closure must be earned, not assumed from the fact of generation.

1. the phases are generated by the same fixed law;
2. the corresponding regimes are genuinely adequate for the phases they explain;
3. the transitions between regimes are conservative over prior history;
4. those transitions are structurally irreducible;
5. no final explanatory stance from the class yields fixed admissible explanatory closure for the whole tower.

A system can generate what it cannot finitely finish explaining.

That slogan marks the place where ordinary complexity talk ends and a new mathematical possibility begins: not merely many states, but lawful generation without final explanatory closure.

What makes the self-transcending generator a new mathematical object is therefore not that it has many states, nor that it surprises bounded observers, but that it occupies a new structural category: exact generation without final explanatory closure. If such objects exist, then explanatory anti-closure is not a symptom of epistemic weakness alone; it is a lawful structural possibility.

8 Paradigm shift as a formal object

One of the deepest aims is to rescue "paradigm shift" from sociological fog and give it a structural backbone. The target is not arbitrary language change, nor a destructive rewrite of the past, nor metaphorical "revolution." It is a disciplined pattern: non-destructive transitions that nonetheless force genuine novelty in what can be explained and certified.

8.1 Conservative extension

The successor regime R_{n+1} must respect what R_n already secured on the available history H_n. Conservativity is the guarantee that the revolution is not vandalism: prior adequacy on certified past structure is preserved when passing upward. Without that constraint, the notion collapses into mere replacement; with it, novelty must squeeze through a narrow gate—it must arrive *despite* honest retention of what came before.

8.2 Irreducibility

At the same time, R_{n+1} must not be a notational variant of R_n for the new work at stake. There must be generated structure that R_n cannot adequately treat while R_{n+1} can. Irreducibility is the mathematical content of "genuinely new explanatory structure": not convenience, but loss of collapse into the prior explanatory basis for the relevant claims.

8.3 Why this is paradigm shift, not metaphor

This is why the theorem package around infinite conservative paradigm towers is not a decorative extension of the theory but one of its deepest results. It shows that explanatory revolution need not be either destructive or merely sociological. It can be historically conservative, mathematically necessary, and still irreducible to any fixed earlier admissible regime. In that sense, paradigm shift becomes a structural theorem rather than a metaphor.

Formally, a paradigm shift here is a transition from R_n to R_{n+1} over H_n such that:

1. $R_n < R_{n+1}$;
2. $\mathsf{ConservativeOver}(H_n, R_n, R_{n+1})$;
3. there exists a new generated phase P_{n+1} with
$$\neg\mathsf{Adeq}(R_n, P_{n+1}) \wedge \mathsf{Adeq}(R_{n+1}, P_{n+1});$$
4. the new regime is not reducible to the old one with respect to the relevant structural content.

Remark 8.1 (Minimality of conservativity in the packaged definition)**.** In the library, the sealed `ParadigmShift` record is equivalent to a *weak* paradigm step together with `ConservativeOver` on the same history (`NoveltyTheory.Core.MinimalHypotheses.paradigmShift_iff_weak_and_conservative`). There are regimes and histories with the weak conjuncts but *without* conservativity, hence without a packaged shift (`NoveltyTheory.Ridge.ConservativeSeparationCountermodel.countermodel_without_history_conservativity`).

9 The first summit: fixed generation without final explanatory closure

The first summit proves that the explanatory tower is never uniformly absorbed by any single final explanatory stance from the admissible class—equivalently, fixed admissible explanatory closure fails for the tower.

$$\exists G\, \exists (P_n, R_n)_{n\in\mathbb{N}} \left[\forall n\, \mathsf{GeneratedBy}(G, P_n) \wedge \forall E \in \mathsf{Adm}\, \exists n\, \neg\mathsf{Expl}(E, G, P_n, R_n)\right].$$

The system remains perfectly lawful; the law generates the whole trace. What fails is final explanation—not because forecast is hard within a fixed architecture, but because *no* architecture of the admissible kind is final across the tower.

This is the first point at which the theory reveals itself as a theory of explanatory anti-closure rather than merely a theory of complicated systems. The point is not that some stance happens to fail. It is that every reducer in the class is defeated somewhere. That is the first true rupture with the hidden creed.

The first summit

A fixed finitely specified generator can produce every relevant phase causally, and yet no fixed admissible explanatory regime is final for the whole generated future.

10 The second summit: no final explanatory stance

The no-final-explanatory-stance form is stronger than "some representation wears out."

> There is no admissible explanatory stance of the relevant type that is final for the generated future.

Finality itself becomes impossible in the specified class—not a local glitch but explanatory anti-closure:

$$\text{fixed law} \not\Rightarrow \text{a final explanatory stance for the generated future.}$$

This is already a very strong claim. It is still not the crown.

The significance of this theorem is not merely technical. It blocks a familiar escape route: the thought that perhaps there is still, somewhere in the architecture, a final explanatory stance from which the whole future story can be closed. In the relevant formal setting, there is not. The failure is not local but architectural.

11 The generalized final crown package

The flagship mathematical export is the **generalized final crown**: a coordinated theorem package in which the central philosophical inversion becomes fully formal. What preceding summit layers establish existentially, the generalized crown stabilizes across disciplined transport and scoped admissible structure. Read as one backbone rather than as isolated peaks, the package includes, in order of conceptual dependence:

1. **Generalized iterated structural ascent.** Strict proof-theoretic ascent for structurally certified sentences along the packaged regimes.
2. **Generalized no fixed structural stratum.** No fixed structural stratum of the relevant formal class is final for the generated future—an anti-closure claim stronger than a one-step gap.
3. **Generalized future defeat of terminality.** Would-be terminality-style universal bounds are defeated in a future-directed, packaged sense for the canonical tower.
4. **Generalized generator-truth family.** Ascent is tied to truths about the generator itself, not to arbitrary propositional noise.
5. **Generalized class transfer.** The phenomenon is realized not only in one witness but across aligned carriers in a nontrivial admissible class.
6. **Generalized non-single-model artifact.** The gap pattern is certified *not* to be merely an artifact of one model, one encoding, or one carrier.

Taken together, these clauses do not merely strengthen antecedent crown packaging. They change its status. What first appears as a dramatic existence result is then shown to persist across a disciplined admissible class. That is what turns the phenomenon from a striking witness into the beginning of a theory.

The crown phenomenon would be dramatically weaker if it held only in one hand-picked model. Transfer is what blocks the cheap escape: "this is a trick of the presentation." When the same structural failure class recurs under disciplined transport while traces remain genuinely distinct in the relevant sense, the philosophical reading strengthens: one is not watching a coding accident but a reproducible pattern of lawful explanatory anti-closure.

The `Bool`$\times\mathbb{N}$ twin construction remains the canonical *existential* staging ground for **class transfer** in the summit package, because it supplies two concrete aligned generators with trace-level mismatch. The naturality-aware theorems in Table 2 then push the *same* output-enum witness `ProvesAtCarrier` gap to **every** stage and every `NaturalAdmissibleInstance` (including $\mathbb{N}$-observation carriers), and record organization-scale and anti-closure consequences under the predicates named there.

Remark 11.1 (Six crown clauses in LEAN 4)**.** Each clause above has a named certificate in `GeneralizedFinalCrownPackage`:

- **Iterated structural ascent.**

  `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_iterated_structural_ascent`

- **No fixed structural stratum** (for every bound B).

  `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_no_fixed_structural_stratum`

- **Future defeat of terminality.**

  `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_future_defeat_of_terminality`

- **Generator-truth family paired with ascent.**

  `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_generator_truth_family`

- **Class transfer.**

  `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_class_transfer`

- **Non-single-model artifact.**

  `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_not_model_artifact`

Remark 11.2 (Natural class transport)**.** Without enlarging quantifiers illegally beyond their formal scopes, the same architecture yields:

(i) **Weak naturality / adequacy transport.**

  `NoveltyTheory.Foundation.NaturalityFacts.naturality_adequacy_transport`

(ii) **Output-enum strict gaps on every aligned `NaturalAdmissibleInstance`.**

  `NoveltyTheory.Ridge.BroadTransfer.naturalClass_preserves_structural_ascent`

  `NoveltyTheory.Ridge.BroadTransfer.broad_transfer_of_generalized_crown`

(iii) **Obstructions for `NaturalBoundedOrganization`.**

`NoveltyTheory.Ridge.UnboundedOrganization.natural_no_final_organization_theorem`

(iv) **Packaged anti-closure on `NaturalNumericSystem`.**

`NoveltyTheory.Ridge.NaturalAntiClosure.natural_anti_closure_theorem`

Informal "inevitability" claims must match these statements' hypotheses.

For witness systems, proof sketches, and LEAN 4 theorem anchors keyed to the crown package above, see Section A.

12 The crown summit: upward explanatory necessity

This section is the central summit of the argument.

If there is a single theoremic-philosophical center of gravity, it is here. Preceding summits establish that fixed lawful generation need not yield final explanatory closure. This summit goes further: it shows that later generated regimes can become necessary, not merely for later phases, but for structural truths about the generator itself. **This is the inversion theorem.**

The prior summits establish explanatory anti-closure: no fixed reducer in the relevant class closes the tower; no final explanatory stance of the relevant type settles the future story. What follows is the deeper inversion. Later generated explanatory regimes can become *necessary* for structural truths about the generator itself. The causal order remains unchanged; what changes is explanatory dependence.

Why it sounds contradictory. If the generator is the source of everything downstream, it is tempting to treat everything "real" about the process as already implicit in the source description. On that hearing, the idea that *later* strata are indispensable for understanding the *earlier source* feels like logical nonsense—as if the effect somehow authored the cause. That hearing conflates generation with explanation.

Why it is not contradictory once the relations are separated. Generation answers: what states arise, in what order, under which fixed update? Explanation answers: which admissible regime is adequate for which structural truths, and what can be proved or expressed at which depth? Those are different questions. Nothing in the forward causal order is reversed when a later regime is required to *state or prove* what an earlier regime could not. The source still generates the whole trace; the derivative layers remain causally downstream; what changes is where certain truths become visible, expressible, or provable.

Why this flips reductionism rather than merely limiting it. A modest limitation story says: "we lack a single brilliant final theory because we are finite." The crown says more. It says the hierarchy of intelligibility need not admit a final admissible stratum *even in principle* relative to the class, and that later strata can be *unavoidable* for truths about the source. That is not a gap in our patience; it is a structural feature of the explanatory relation under lawful generation.

The usual picture runs like this:

1. the generator is fundamental;
2. the generated layers are derivative;
3. therefore the generated layers cannot be strictly necessary for understanding the generator except as convenience.

The crown says step (3) can fail.

$$\exists G, \exists (R_n), \exists (\Phi_n)\ \forall n,\ \Phi_n \text{ is a structural truth about } G \wedge \mathsf{ProvesAt}(R_{n+1}, \Phi_n) \wedge \neg \mathsf{ProvesAt}(R_n, \Phi_n).$$

Read verbally, the theorem says: for each stage there is a structural truth about the generator that becomes provable only from a later regime. The generator remains the source of the whole tower, but the tower it generates becomes necessary for understanding the generator itself.

The claim is not about later outputs alone. It is about structural truths of the source process: the derivative becomes necessary for understanding the source.

In the formalization, the schematic Φ_n can be taken concretely from the retro-structural family `histSeqUpto` n—a finite diagonal history of trace-equalities (i, i) for $i < n + 1$—which is `IsStructuralSentenceV2` and satisfies `ProvesAt` $(n{+}1)\, \Phi_n \wedge \neg$`ProvesAt` $n\, \Phi_n$. The named certificates for this family and for generator-anchored sentences such as `traceEq` $n\, n$ are recorded in Table 1 and Section A (retro-structural family, strict gap at successor depth, generator-truth packaging).

The crown inversion

Later generated explanatory regimes can become necessary for structural truths about the generator itself.

13 Explanatory order and causal order

This section does three jobs at once.

(1) Formal separation. Causal order is the forward structure of lawful production:

$$s_0 \mapsto s_1 \mapsto s_2 \mapsto \cdots$$

Explanatory order tracks where structural truths become expressible, adequate, or provable within the theory's fixed explanatory regimes. The two orders can diverge: a fact can be causally "early" while remaining explanatorily "late."

(2) No backward causation. Upward explanatory necessity does not mean later regimes pull levers in the past. Causal sufficiency of the generator and forward dynamics are untouched. The inversion concerns intelligibility and proof-theoretic access, not forces traveling up the time arrow.

(3) Upward dependence without causal reversal. Explanatory dependence can run from higher strata back toward truths about the generator *without* reversing causal priority. Think of it as a second coordinate system laid over the same lawful trajectory: the causal coordinate orders production; the explanatory coordinate orders what must be said to honor the structure. They need not coincide.

Explanatory order need not mirror causal order.

This is the opening for a new conception of explanation. Causal order may still run downward or forward from a fixed source, while explanatory dependence climbs upward through generated strata that the source itself necessitates but does not finitely close. That is the formal core of lawful self-transcendence.

14 Retroactive structural revelation

Later regimes need not merely forecast the next segment of trace. Once explanatory order splits from causal order, a sharper phenomenon appears: later regimes can *disclose latent structure in earlier history*—truths about $P_{\le n}$ that were not expressible, provable, or structurally visible from R_n, yet become available from R_m for $m > n$.

$$\exists n < m, \exists \Psi \left[R_m \vdash \Psi(P_{\le n}) \wedge R_n \nvdash \Psi(P_{\le n}) \right].$$

Retroactive structural revelation is therefore not an incidental byproduct of the theory. It is one of its deepest consequences. Explanation grows not only by extending into the future but by reorganizing the intelligibility of the past. Later regimes do not merely forecast what comes next; they can disclose what had already been there without yet being structurally sayable. The certified template sentences `histSeqUpto` n are exactly of this "initial segment made explicit" form; see Table 1 and the retro discussion in Section A.

15 Why this is not chaos, not computational irreducibility, not emergence rhetoric, and not mere bookkeeping

Disagreement coordinates: reject these identifications

The target is *not* chaos, opaque simulation cost, loose emergence talk, or mere bookkeeping. Four distinct failures of identification:

- **Generation $\neq$ explanation.** Lawful production of a phase does not entail its explanatory reducibility to any fixed regime drawn from the relevant class.
- **Simulation $\neq$ explanatory reduction.** Stepping the dynamics can outrun explanatory collapse: exact generation does not hand you final understanding in the packaged sense.
- **Observational sameness $\neq$ explanatory sameness.** Matching coarse traces does not settle adequacy for structurally certified truths at issue in the theory.
- **Complexity growth $\neq$ explanatory anti-closure.** "Getting richer" in an informal sense does not by itself imply the systematic failure of every fixed final reducer in the class.

15.1 Chaos

Chaos: sensitive dependence and unstable long-term prediction *within* a fixed explanatory architecture. Novelty theory: that architecture, as constrained by the relevant class, may not be final across the tower at all. Difficulty $\neq$ systematic explanatory anti-closure.

15.2 Not computational irreducibility

Computational irreducibility [8] concerns the absence of substantial shortcuts for computing later states of a process. The present theory concerns something stronger and different: explanatory anti-closure—even for a fully lawful generated trace, no single regime need constitute a final explanatory stance in the relevant sense. One may have exact generation, exact simulation, or observational agreement and still lack explanatory reduction in the relevant formal sense. The issue here is not merely that the future is costly to compute; it is that explanation itself need not close under a fixed explanatory stance drawn from the class.

15.3 Not generic emergence rhetoric

The framework does not rest on loose "emergence," "levels," or "useful descriptions." It rests on reductions licensed by explicit predicates, adequacy, conservative shifts, generator-anchored truths, and proof-theoretic ascent—where later structure can be mathematically necessary and not uniformly collapsible to any fixed prior explanatory stance of the relevant class.

15.4 Why this is not just Gödel or Tarski restated

The theory is adjacent to the great limitative results of logic, but it is not a repackaging of them. Gödelian incompleteness [3] concerns the inability of a sufficiently expressive formal system to capture all truths of a certain kind within one proof system. Tarskian undefinability [6] concerns the impossibility of defining truth for a language wholly within that same language under the relevant formal conditions. Both are canonical results about truth, proof, and definability in highly general formal settings.

The present theory targets a different object. Its central concern is not truth in arithmetic as such, nor truth-predicates for formal languages as such, but final explanatory closure for generated phase and regime towers under a disciplined class of reducers. The question is not simply whether some truths outrun one proof system, but whether fixed lawful generation forces any fixed final explanatory stance for what that law itself produces. The answer proved here is no.

There is nevertheless a real family resemblance. In both traditions one encounters a failure of final explanatory closure from a single fixed internal apparatus; principled limits on closure; and surfaces that look "fully governed from within" yet are not exhaustible from within by one fixed apparatus. But the mechanism, target, and philosophical interpretation differ. The object here is lawful self-transcending explanatory architecture: generated phases, explanatory regimes, conservative but irreducible succession, and, at the crown, the necessity of later regimes for structural truths about the generator itself.

So the correct comparison is not identity but analogy of shape. The theory belongs in the same broad family of anti-closure results, but it is not simply Gödel or Tarski with new terminology. Its novelty lies in making explanatory anti-closure under exact generation into a formal object in its own right.

15.5 Why this is not merely Kuhnian paradigm-shift talk

Kuhn [4] made vivid that scientific development is not always cumulative in a simple linear sense, and that transitions between paradigms can reorganize what counts as problem, evidence, and explanation. That historical and philosophical insight is important. But it is not yet a formal theorem of any particular mathematical structure.

The notion of paradigm shift used here is narrower, sharper, and explicitly structural. A paradigm shift in this theory is not a sociological episode, not a psychological conversion, and not merely a change in vocabulary. It is a formally constrained transition between explanatory regimes: conservative over prior certified history, yet irreducible with respect to the relevant generated phase and reduction class. The point is not that communities change their minds. The point is that there can exist lawful generative systems for which such conservative but irreducible explanatory succession is mathematically forced.

That difference matters. Kuhnian language by itself leaves open whether paradigm shifts are contingent features of scientific communities, temporary epistemic limitations, or convenient redescriptions of a deeper fixed theory. The present theory addresses a different question: whether, under explicit formal conditions, explanatory anti-closure and conservative irreducible succession can be structurally real. The result is therefore not a formalization of Kuhn's history.

It is a theoremic framework that makes one important aspect of "paradigm shift" precise enough to prove.

15.6 Not "merely engineered" admissibility

A common informal reply is that the crown depends on *specially chosen* predicates or coding. Two distinctions block a quick dismissal. First, explicit **obstruction lemmas** rule out illegitimate universal diagonal maneuvers over raw interfaces: the trust boundary is written into the formal predicates, not hidden. Second, the **naturality-aware block** (Section 11, Table 2) separates **weak naturality / adequacy transport** from **existence of a crown package**, carries the output-enum `ProvesAtCarrier` gap across all aligned `NaturalAdmissibleInstance` carriers (not only `Bool` $\times\mathbb{N}$), and states organization / anti-closure consequences under `NaturalBoundedOrganization` and `NaturalNumericSystem`. None of this asserts a universal over all deterministic systems; each theorem fixes the quantifiers its statement actually uses.

15.7 Not "open-ended inquiry" in the everyday sense

Field manuals already preach intellectual humility: keep models provisional, expect surprises, distinguish data from interpretation. Those ethical norms are compatible with *belief* in a final true theory "somewhere out there," or with the *hope* that some imagined limit vocabulary would close the explanatory ledger if only it were reachable. The present theorems concern a different conditional. They display *constructed* lawful systems in which forward generation is complete for the tower at issue and yet *no reducer from the named class* is final—not because investigators lack patience, but because the failure is *proved* for every candidate closure interface in that class, under the stated definitions. Distinguishing patience from provable non-closure is exactly what stops the crown from collapsing into a sermon everyone already endorses.

16 Why this is not just genotype/phenotype

Genotype/phenotype is a one-step manifestation gap. This theory is an unbounded tower with conservative irreducible succession and upward necessity at the crown. The analogy can at best illuminate the foothills. It cannot capture the theoremic heart of the argument, which is a multi-stage, formally disciplined, upwardly necessary architecture of explanatory non-closure.

17 Lawful self-transcendence as a third philosophical option

Three stances (interpretive)

Naïve reductionism. Everything generated from below, therefore finally explainable from below.

Mystification. If not reducible, then outside law or science.

Third option. Lawful self-transcending structure—fully generated, fully lawful, not finitely explanation-closed (relative to the admissible class at issue), with possible upward necessity as in Section 12.

Why the mathematics matters. The framework supplies a rigorous language for separating mere complexity, mere redescription, genuine structural novelty, true explanatory phase shifts, and the *non-collapse* of explanation under exact generation. Without such distinctions, debates about reduction, emergence, and explanation talk past one another. The point is not only that some formal possibility exists; it is that one can *name* and *prove* a coherent profile of lawful explanatory anti-closure strong enough to break the hidden creed without breaking naturalism.

If this framework is right, then the deepest lesson is not merely that the future may surprise us, nor that science may need new theories. It is that full lawfulness and full generative completeness do not force final explanatory closure. The space of lawful systems is wider than reductionism assumed. That is why lawful self-transcendence is not only a new mathematical object, but a new philosophical possibility.

A **naturality-aware** extension fixes weak principles linking adequacy across stages, checks structural sentence families against them, and transports output-enum ascent across the whole `NaturalAdmissibleInstance` class together with organization and anti-closure consequences on explicit natural predicates. The formal hypotheses are part of the content, not an afterthought.

18 What is formally proved, and what lies outside scope

Trust boundary (scope at a glance)

Inside scope (formal). Packaged lawful generators with iterated structural ascent, failure of fixed final admissible closure, class transfer, generator-truth alignment, naturality-aware transport under the predicates named in Table 2.

Outside scope. "Every deterministic system," backward causation, universal raw-interface diagonal, infinite paradigm tower for every regime family, universal `FutureDefeat` over all lawful numeric generators—unless a cited lemma states otherwise (`NoveltyTheory.Ridge.UniversalUpwardNecessity.upward_necessity_universal_obstruction`).

Reader discipline. Treat any informal headline as provisional until it matches a row in Table 1, Table 2, or Table 3.

The paper's mathematical claims are strong but sharply scoped. In the formalized models, there exist lawful generators with iterated structural ascent, failure of final explanatory closure relative to the admissible class, future defeat of terminality in the packaged crown sense, generator-anchored truth families, class transfer across a nontrivial family of aligned carriers (including concrete $\texttt{Bool}\times\mathbb{N}$ witnesses), and non-single-model-artifact results.

A further family of theorems (summarized in Table 2) gives **broad transfer** of the output-enum carrier gap for all aligned `NaturalAdmissibleInstance` values, **bounded natural organization** obstructions for the standard future predicate, and **natural anti-closure** on `NaturalNumericSystem`—each under the hypotheses named in the corresponding Lean 4 declarations.

What the formal package does not entail is equally important. It does not imply that every deterministic system exhibits the crown phenomenon, that every conceivable explanatory notion fails globally, that every organization must be infinite, or that causation runs backward. It does not imply a universal diagonal over all raw admissible interfaces, nor an infinite paradigm tower for every singleton-within regime family. Those literal universal glosses are false under the formal predicates where explicit obstruction applies.

This boundary is not a retreat. It is the condition of the theory's strength. The scoped theorem family is already sufficient to establish the existence, structure, and transfer of lawful non-final explanation under fixed law; the naturality-aware block adds rigorously scoped claims that undercut a narrow "presentation trick" dismissal.

19 Conclusion

Big payoffs

1. The tacit creed fails. Fixed fundamental law does *not* entail a fixed final explanatory stance for everything important about the generated future in the relevant class: there are rigorous models where the causal trace is complete and final explanatory closure is not.

2. A whole tower, not a glitch. The phenomenon is packaged as explanatory anti-closure across phases and regimes, conservative yet irreducible paradigm succession, and (under the cited predicates) obstruction of would-be terminal organization—not a vocabulary shortfall or a one-off surprise.

3. Crown inversion. Later regimes can be unavoidable for structural truths about the source. Explainability and proof-theoretic access need not track causal priority; upward explanatory dependence is compatible with forward generation.

4. Transfer, not a single encoding. Class transfer and the naturality-aware block show the flagship gap pattern and companion obstructions are stable across aligned carriers and explicit natural hypotheses—not a trick of one hand-picked model (Section 11, Table 2).

5. Honest quantifiers + machine proof. Scope is exactly what the formal statements assert; the formal development cited here is machine-checked end to end (Table 1–Table 3, Section A).

6. Minimality and numeric sharpness land in the library. Conservativity can fail while weak paradigm-step conjuncts still hold; numeric `FutureDefeat` aligns with trace dichotomies and admits neither unrestricted universal generalization nor silent strengthening of hypotheses without new theorems (Table 3).

The hidden creed with which the argument began can no longer be treated as self-evident. Fixed law does not imply a final explanatory stance.

In the proved models, a single lawful generator can produce an infinite tower of phases and explanatory regimes while defeating every candidate for fixed admissible explanatory closure of the relevant class. The succession can be conservative without being reducible, historically preserving what came before while forcing genuinely new explanatory structure. At the crown, later generated regimes can become necessary for structural truths about the generator itself.

That conjunction is the paper's real theoremic contribution: exact generation without final explanatory closure—stronger than complexity growth, informal unpredictability, loose emergence talk, or a one-step manifestation gap.

Explanation need not bottom out where causation bottoms out: a system may be fully lawful and fully generative without being finitely explanation-closed.

If that possibility is genuinely formal, then a new mathematics begins here: not merely a mathematics of what fixed laws generate, but a mathematics of non-final explanation under fixed law.

Under the hypotheses named in Table 2, the witness-level ascent is stable across aligned observation encodings, and organization-scale anti-closure is available on the corresponding predicates—so the crown is not a single-carrier quirk.

Landmark criterion (sober). A paper of this kind earns attention not by pretending that no one noticed science changes its mind, but by showing that a *precise* configuration—lawful totality of the trace, systematic explanatory anti-closure for a whole class of closure candidates, conservative irreducible succession, crown inversion, and disciplined transport where proved—was not available as folklore. The formalization supplies the antidote to two symmetric mistakes:

smuggling universal final closure in with lawfulness, and mistaking the theorems for tautologies everyone already knew. The point, throughout, is to keep the quantifiers honest.

A Witness systems, proof sketches, and Lean 4 anchors

Code availability. Lean 4 library `novelty-theory-lean` (`NoveltyTheory/`). Build: `lake build`. Summit certificates cited in the main text are checked in that source (see tables below for declarations and LEAN 4 names).

This appendix lists formal witnesses, the principal LEAN 4 certificates, and compact proof sketches. Gray **At a glance** blocks summarize each witness; blue **Proof sketch** boxes isolate one-line arguments where useful.

A.1 Generative systems and the canonical counter

At a glance. A discrete-time deterministic generator emits an infinite observable trace. The default dynamical witness is a bare counter on $\mathbb{N}$ whose trace is the identity.

- **Formal generator.** State S, observations X, parameters $(s_0, \tau, \mathrm{out})$, trace $\mathrm{trace}_G(n) = \mathrm{out}(\tau^n(s_0))$.
- **Lean 4 structure.** `NoveltyTheory.Core.GenerativeSystem`
- **Canonical counter `natCounter`.** $S = X = \mathbb{N}$, $s_0 = 0$, $\tau =$ `Nat.succ`, $\mathrm{out} = \mathrm{id}$; hence every n is observed at time n.
  - Definition: `NoveltyTheory.Models.SignatureTower.natCounter`
  - Trace: `NoveltyTheory.Models.SignatureTower.natCounter_trace`
  - Singleton phases `phaseSingleton` k are generated: `NoveltyTheory.Models.SignatureTower.natCounter_generated`

A.2 Sentences and depth-indexed derivability

At a glance. Truth `HoldsAt natCounter` and derivability `ProvesAt` m are defined by recursion on the shape of a `Sentence`. The critical gate for crown witnesses is a uniform inequality tying *list content* to *proof depth*.

- **Depth calculus.** Base depth calculus: `NoveltyTheory.Models.InvariantTower.provesAtDepth`. Sentence-level derivability: `NoveltyTheory.Models.SentenceProvability.ProvesAt`.
- **Example clause (`outputEnumMem` $\ell\, x$).** Requires:

(a) $x \in \ell$ (the named output appears in the finite list);

(b) $\forall y \in \ell,\ y \leq m$ (no list element exceeds the *current* depth m);

(c) a depth-m certificate for the singleton-phase atom at x.

Raising m is what allows longer lists to become uniformly licensable: later depth unlocks stricter global bounds on ℓ.

A.3 The output-enumeration gap (transferable witness)

At a glance. Use the list $\{0, \ldots, n+1\}$ and ask about membership of n. The sentence is *true* on `natCounter` at every stage, but derivability at depth n fails because $n+1 \in \ell$ breaks the "all entries $\leq n$" side condition. One more step of depth repairs the bound.

Definition (paper abbreviation). $\textsc{OutEnum}_n$:= `outputEnumMem (List.range (n+2))` n.

$$\texttt{ProvesAt}\,(n+1)\,\textsc{OutEnum}_n \;\wedge\; \neg\texttt{ProvesAt}\,n\,\textsc{OutEnum}_n$$

- **Main certificate.** `NoveltyTheory.Models.OutputEnumCrownFamily.outputEnumCrownWitness_proves_succ_not_at`
- **List / bound lemmas.**
  - `NoveltyTheory.Models.OutputEnumCrownFamily.mem_range_self_lt_add_two`
  - `NoveltyTheory.Models.OutputEnumCrownFamily.forall_mem_range_le_succ_self`
- **Structural status / non-collapse.**
  - `NoveltyTheory.Models.OutputEnumCrownFamily.isStructural_outputEnumCrownWitness`
  - `NoveltyTheory.Models.OutputEnumCrownFamily.outputEnumMem_not_geOutput_abbrev`
- **Carrier-general gap (any aligned section).** For an arbitrary observation type X and `CarrierSection` X, the embedded output-enum witness has the same strict `ProvesAtCarrier` profile: `NoveltyTheory.Foundation.PhaseGeneralizationFacts.carrier_outputEnum_gap_at`.
- **Broad natural class.** For every `NaturalAdmissibleInstance` (definitionally: aligned generator + section + trace alignment), the same pattern holds at every stage. Packaged conjunction and local lemma:
  - `NoveltyTheory.Ridge.BroadTransfer.broad_transfer_of_generalized_crown`
  - `NoveltyTheory.Ridge.BroadTransfer.naturalClass_preserves_structural_ascent`

Proof sketch.

1. **At $m = n+1$.** Check $n \in \texttt{range}(n+2)$, verify $\forall y \in \ell,\ y \leq n+1$, supply the phase atom at n (uses `upward_phase_derivability_gap`).
2. **At $m = n$.** Assume a proof; instantiate the uniform bound at $y = n+1 \in \ell$ to force $n+1 \leq n$, contradiction.

A.4 Retro-structural ascent (`histSeqUpto`)

At a glance. A second crown family bundles many earlier diagonal trace-equalities (i, i) for $i < n{+}1$. It is certified retro-structural (`IsStructuralSentenceV2`) and satisfies the same strict `ProvesAt` gap by a diagonal counting argument at depth.

- **Definition.** `NoveltyTheory.Foundation.RetroStructuralTruthV2.histSeqUpto`
- **Strict ascent.** `NoveltyTheory.Ridge.RetroStructuralGap.histSeqUpto_proves_succ_not_at`
- **Uniform family packaging.** `NoveltyTheory.Foundation.StructuralSentenceHierarchyV2.structuralV2_crown_family`
- **Summit re-export.** `NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_iterated_structural_ascent`

Proof sketch. At $n+1$, every pair in the finite `histSeq` list has time index $< n+1$, so each diagonal atom is an eligible fringe proof. At n, the pair (n, n) is still in the list but the diagonal trace atom `traceEq` $n\,n$ is not derivable at depth n (`NoveltyTheory.Models.InvariantTower.not_proves_trace_diag`).

A.5 Future defeat of terminality

At a glance. `FutureDefeat` G: for every proposed output bound, some later time exceeds it. For `natCounter`, take $t =$ `bound` $+ 1$.

- **Predicate.** `NoveltyTheory.Models.StratifiedFinality.FutureDefeat`
- **Closed proof.**
  - `NoveltyTheory.Models.StratifiedFinality.natCounter_futureDefeat`
  - Uses trace: `NoveltyTheory.Models.SignatureTower.natCounter_trace`
- **Sharp numeric boundary (no extra crown assumptions).** On observation type $\mathbb{N}$, failure of `FutureDefeat` is equivalent to a uniformly bounded numeric trace; dichotomy: `NoveltyTheory.Ridge.ClosureCollapseBoundary.futureDefeat_or_existsBoundedNumericTrace`.
- **Not universal over all generators.** Constant observable output obstructs `FutureDefeat` for a lawful system on $\mathbb{N}$; hence `NoveltyTheory.Ridge.UniversalUpwardNecessity.upward_necessity_universal_obstruction`.

A.6 Paradigm packaging: conservativity is not decorative

At a glance. A *packaged* paradigm shift layers history conservativity on top of a strictly weaker "weak step" record. Dropping conservativity can leave all three weak conjuncts while still failing the full package.

- **Equivalence.** `NoveltyTheory.Core.MinimalHypotheses.paradigmShift_iff_weak_and_conservative`
- **Separation witness.** `NoveltyTheory.Ridge.ConservativeSeparationCountermodel.countermodel_without_history_conservativity`

Remark A.1. The witness uses singleton phases on $\mathbb{N}$ and differing output-set adequacy predicates; it is meant to pin a *definition-theoretic* point, not to model physical dynamics.

A.7 No fixed structural stratum (organization height)

At a glance. Fix a finite organizational height B. Some `natCounter`-generated phase is invisible to the corresponding finite signature stratum, and no abstract height-B organization can *totalize* the standard future ordering on stages.

- **Packaged obstruction.** `NoveltyTheory.Ridge.OrganizationAbstractObstruction.organization_obstruction_supports_generalized_crown`
- **Ingredients (for manual tracing).** `NoveltyTheory.Models.SignatureTower.finite_signature_cannot_organize_full_ladder`, `NoveltyTheory.Ridge.OrganizationHeightObstruction.no_finite_adequate_organization_totalizes_future`

A.8 `Bool`$\times\mathbb{N}$ carriers, alignment, and class transfer

At a glance. Twin laws disagree on a `Bool` tag at every time, yet each twins the counter after embedding. The same output-enumeration gap survives on the carrier calculus (`ProvesAtCarrier`). This symmetric pair is the canonical *existential* witness for **class transfer** in `GeneralizedFinalCrownPackage`; broad transfer across `NaturalAdmissibleInstance` is the natural-class extension recorded below.

Object	Role / reference
`trueBoolProdNat`	Always-`true` tag; trace $n \mapsto (\texttt{true}, n)$. `NoveltyTheory.Ridge.CrownTransfer.trueBoolProdNat`
`altBoolProdNat`	Always-`false` tag; same $\mathbb{N}$ skeleton. `NoveltyTheory.Ridge.CrownTransfer.altBoolProdNat`
`trace_true_vs_alt`	Pointwise inequality on traces. `NoveltyTheory.Ridge.CrownTransfer.trace_true_vs_alt`
Alignment	`true_bool_aligns`, `alt_bool_aligns`. `NoveltyTheory.Ridge.CrownTransfer.true_bool_aligns` `NoveltyTheory.Ridge.CrownTransfer.alt_bool_aligns`
Faithful embed	Pull `natCounter` truth back and forth. `NoveltyTheory.Foundation.PhaseGeneralizationFacts.phase_general_embed_faithful`
Carrier gap	`NoveltyTheory.Ridge.CrownTransfer.true_general_phase_gap_at` (and symmetric `alt_...`)
Class transfer	`NoveltyTheory.Ridge.CrownTransfer.crown_transfers_across_admissible_class`
Non-artifact	`NoveltyTheory.Ridge.CrownTransfer.generalized_crown_not_single_model_artifact`

Proof sketch. `true_general_phase_gap_at`: forward direction refines a purported depth-n carrier proof of the embedded witness to a depth-n proof of the underlying `Sentence` $\mathbb{N}$, contradicting `outputEnumCrownWitness_proves_succ_not_at`. Transfer bundles that gap with an existential witness for `G` and aligned `C`; non-artifact instantiates two concrete aligned generators and the first time mismatch.

A.9 Natural admissible class, broad transfer, and twin encodings

At a glance. `NaturalAdmissibleInstance` packages any observation type α with a generative system, a `CarrierSection` α, and alignment between numeric trace and embedded observations. `carrier_outputEnum_gap_at` implies `naturalClass_preserves_structural_ascent` and the staged conjunction `BroadTransferStatement`. Two `Bool`$\times\mathbb{N}$ instances with differing traces show that the gap is not an encoding artifact *within* that class.

- **Aligned instance.** `NoveltyTheory.Core.NaturalAdmissibleClass.NaturalAdmissibleInstance`
- **Concrete shapes.**
  - `NoveltyTheory.Models.NaturalClassExamples.natCounterAdmissible`
  - `NoveltyTheory.Models.NaturalClassExamples.boolTrueAdmissible`
  - `NoveltyTheory.Models.NaturalClassExamples.boolFalseAdmissible`
- **Broad transfer.** `NoveltyTheory.Ridge.BroadTransfer.broad_transfer_of_generalized_crown`
- **Trace separation + non-artifact.**

- NoveltyTheory.Ridge.BroadTransfer.multiple_nontrivially_distinct_instances
  - NoveltyTheory.Ridge.BroadTransfer.crown_not_encoding_artifact_in_natural_class

- **Organization inevitability layer.** NoveltyTheory.Ridge.UnboundedOrganization.natural_no_final_organization_theorem
- **Anti-closure packaging.** NoveltyTheory.Ridge.NaturalAntiClosure.natural_anti_closure_theorem

A.10 Generator-truth linkage

At a glance. The summit packages *ascending* structural sentences together with *elementary* generator-structural trace facts traceEq $n\,n$ that really hold on natCounter.

- **Theorem.** NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_generator_truth_family
- **Proof idea.** Split the conjunction: the ascent family is structuralV2_crown_family; each traceEq $n\,n$ is generator-structural and true by definition of natCounter.trace.

B Machine-checked certificates and reproducibility

Every theorem statement cited from NoveltyTheory/ is checked in Lean 4 without sorry. Readers can rebuild the library with standard lake workflows; Table 3 and \leanref paths point into that tree. Optional mechanical recheck of this file's Lean path strings: scripts/verify_paper_leanrefs.py.

C Theorem tables

Table 1 lists the six summit declarations of the generalized final crown package. Table 2 lists identifiers for the naturality-aware extension (broad transfer, distinct encodings within the natural class, organization obstruction, anti-closure packaging). Table 3 surveys supporting theorems and definitions in rough dependency order.

Crown clause	Lean 4 identifier
Iterated structural ascent	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_iterated_structural_ascent
No fixed structural stratum (parameter B)	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_no_fixed_structural_stratum
Future defeat of terminality	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_future_defeat_of_terminality
Generator-truth family + ascent	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_generator_truth_family
Class transfer (Bool$\times\mathbb{N}$)	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_class_transfer
Not a single-model artifact	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_not_model_artifact

Table 1: Generalized final crown package: packaged summit theorems.

Natural-class clause	Lean 4 identifier
Broad transfer at every stage (ℕ and `Bool`×ℕ instances)	`NoveltyTheory.Ridge.BroadTransfer.broad_transfer_of_generalized_crown`
Non-encoding artifact in natural class	`NoveltyTheory.Ridge.BroadTransfer.crown_not_encoding_artifact_in_natural_class`
Natural anti-closure (packaged)	`NoveltyTheory.Ridge.NaturalAntiClosure.natural_anti_closure_theorem`

Table 2: Selected identifiers for naturality-aware transport and anti-closure (quantifiers exactly as in the Lean 4 statements).

Table 3: Extended theorem inventory for the generalized crown mathematical spine (**40** curated dependency rows; the live `NoveltyTheory/` library is strictly larger, so this table is not exhaustive). "Depends on" lists immediate named ingredients in the formal proof, not the full transitive closure.

Layer	Lean 4 declaration	Informal content	Immediate dependencies
Core	`NoveltyTheory.Core.GenerativeSystem`	Deterministic generator $(s_0, \tau, \mathrm{out})$ and trace.	`(primitive)`
Models	`NoveltyTheory.Models.SignatureTower.natCounter`	Canonical counter: state ℕ, succ, identity observation.	`NoveltyTheory.Core.GenerativeSystem`
Models	`NoveltyTheory.Models.SignatureTower.natCounter_trace`	$\mathrm{trace}(n) = n$.	induction on n
Models	`NoveltyTheory.Models.SignatureTower.natCounter_generated`	Singleton phase $\{k\}$ generated by `natCounter`.	`NoveltyTheory.Models.SignatureTower.natCounter_trace`
Models	`NoveltyTheory.Models.SentenceProvability.ProvesAt`	Depth-m derivability on `Sentence`.	`NoveltyTheory.Models.InvariantTower.provesAtDepth`
Models	`NoveltyTheory.Models.InvariantTower.not_proves_trace_diag`	Diagonal trace atom not derivable at matching depth.	`NoveltyTheory.Models.InvariantTower`
Models	`NoveltyTheory.Models.InvariantTower.upward_phase_derivability_gap`	Strict inclusion of derivable facts when depth increases.	`NoveltyTheory.Models.InvariantTower`
Models	`NoveltyTheory.Models.OutputEnumCrownFamily.outputEnumCrownWitness_proves_succ_not_at`	`outputEnumMem` list witness: provable at $n+1$, not at n.	range lemmas `NoveltyTheory.Models.InvariantTower.upward_phase_derivability_gap`
Models	`NoveltyTheory.Models.StratifiedFinality.natCounter_futureDefeat`	No uniform bound on `natCounter` trace values (`FutureDefeat`).	`NoveltyTheory.Models.SignatureTower.natCounter_trace`
Ridge	`NoveltyTheory.Ridge.ClosureCollapseBoundary.not_futureDefeat_iff_existsBoundedNumericTrace`	Numeric layer: ¬`FutureDefeat` ↔ uniformly bounded trace.	`NoveltyTheory.Models.StratifiedFinality.FutureDefeat` `NoveltyTheory.Core.ClosureDichotomy.ExistsBoundedNumericTrace`
Ridge	`NoveltyTheory.Ridge.ClosureCollapseBoundary.futureDefeat_or_existsBoundedNumericTrace`	Dichotomy: unbounded recurrence vs. uniform trace bound ($X = \mathbb{N}$).	classical case split
Ridge	`NoveltyTheory.Ridge.UniversalUpwardNecessity.upward_necessity_universal_obstruction`	¬∀ lawful `GenerativeSystem`, `FutureDefeat` (constant trace).	`NoveltyTheory.Ridge.MinimalityCountermodels.not_futureDefeat_of_constant_numeric_trace`

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Table 3 (continued)

Layer	Lean 4 declaration	Informal content	Immediate dependencies
Core	NoveltyTheory.Core.MinimalHypotheses.paradigmShift_iff_weak_and_conservative	ParadigmShift $\leftrightarrow$ weak step $\wedge$ ConservativeOver.	definitions
Ridge	NoveltyTheory.Ridge.ConservativeSeparationCountermodel.countermodel_without_history_conservativity	Inhabited weak paradigm step without history conservativity.	output-set adequacy on phaseSingleton
Summits	NoveltyTheory.Summits.ClosureDichotomySummit.closure_dichotomy_numeric_trace	Summit packaging: numeric FutureDefeat $\vee$ bounded-trace disjunction.	NoveltyTheory.Ridge.ClosureCollapseBoundary.futureDefeat_or_existsBoundedNumericTrace
Summits	NoveltyTheory.Summits.UniversalUpwardNecessitySummit.upward_necessity_universal_obstruction	Summit re-export: universal FutureDefeat obstruction.	NoveltyTheory.Ridge.UniversalUpwardNecessity.upward_necessity_universal_obstruction
Models	NoveltyTheory.Models.SignatureTower.finite_signature_cannot_organize_full_ladder	Fixed signature height misses some singleton phase.	NoveltyTheory.Models.SignatureTower.tower_phase_not_explained_by_fixed_regime
Fdn.	NoveltyTheory.Foundation.RetroStructuralTruthV2.histSeqUpto	Initial-segment diagonal histSeq sentence.	NoveltyTheory.Core.Sentence
Fdn.	NoveltyTheory.Foundation.RetroStructuralTruthV2.isRetroStructuralV2_histSeqUpto	histSeqUpto is retro-structural V2.	list/range facts
Ridge	NoveltyTheory.Ridge.RetroStructuralGap.histSeqUpto_proves_succ_not_at	Same strict ProvesAt gap for histSeqUpto.	NoveltyTheory.Ridge.RetroStructuralGap.histSeqUpto_proves_succ NoveltyTheory.Ridge.RetroStructuralGap.histSeqUpto_not_proves_at
Ridge	NoveltyTheory.Ridge.RetroStructuralGap.histSeqUpto_not_proves_at	Failure at n: encodes histSeqUpto_mem_diag + diagonal obstruction.	NoveltyTheory.Models.InvariantTower.not_proves_trace_diag
Fdn.	NoveltyTheory.Foundation.StructuralSentenceHierarchyV2.structuralV2_crown_family	Φ_n = histSeqUpto n with IsStructuralSentenceV2 ascent.	NoveltyTheory.Foundation.RetroStructuralTruthV2.isRetroStructuralV2_histSeqUpto NoveltyTheory.Ridge.RetroStructuralGap.histSeqUpto_proves_succ_not_at
Ridge	NoveltyTheory.Ridge.OrganizationHeightObstruction.no_finite_adequate_organization_totalizes_future	Finite-height org cannot realize all nested future stage pairs.	bound B vs. $B+1$ on stages
Ridge	NoveltyTheory.Ridge.OrganizationHeightObstruction.organization_height_obstruction	Signature miss + abstract height obstruction (scoped OrganizationV2).	NoveltyTheory.Models.SignatureTower.finite_signature_cannot_organize_full_ladder NoveltyTheory.Ridge.OrganizationHeightObstruction.no_finite_adequate_organization_totalizes_future

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Table 3 (continued)

Layer	Lean 4 declaration	Informal content	Immediate dependencies
Ridge	NoveltyTheory.Ridge.OrganizationAbstractObstruction.organization_obstruction_supports_generalized_crown	Same pattern for AdmissibleBoundedOrganization.	NoveltyTheory.Models.SignatureTower.finite_signature_cannot_organize_full_ladder NoveltyTheory.Ridge.OrganizationHeightObstruction.no_finite_adequate_organization_totalizes_future
Fdn.	NoveltyTheory.Foundation.PhaseGeneralizationFacts.phase_general_embed_faithful	natCounter truth $\leftrightarrow$ aligned carrier truth after embed.	NoveltyTheory.Core.GeneralizedCarrier.AlignsCarrier
Ridge	NoveltyTheory.Ridge.CrownTransfer.trueBoolProdNat NoveltyTheory.Ridge.CrownTransfer.altBoolProdNat	Twin generators on Bool$\times\mathbb{N}$; traces differ on Bool.	NoveltyTheory.Core.GenerativeSystem
Ridge	NoveltyTheory.Ridge.CrownTransfer.true_general_phase_gap_at	Carrier-side ProvesAtCarrier gap for embedded enum witness.	NoveltyTheory.Models.OutputEnumCrownFamily.outputEnumCrownWitness_proves_succ_not_at embed equations
Ridge	NoveltyTheory.Ridge.CrownTransfer.crown_transfers_across_admissible_class	$\exists$ aligned Bool$\times\mathbb{N}$ system with gap + trace-inequality class-wide.	NoveltyTheory.Ridge.CrownTransfer.true_general_phase_gap_at NoveltyTheory.Ridge.CrownTransfer.trace_true_vs_alt
Ridge	NoveltyTheory.Ridge.CrownTransfer.generalized_crown_not_single_model_artifact	Two aligned generators with distinct traces.	NoveltyTheory.Ridge.CrownTransfer.trueBoolProdNat NoveltyTheory.Ridge.CrownTransfer.altBoolProdNat alignment lemmas
Summits	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_iterated_structural_ascent	Packaged structural V2 strict ascent.	NoveltyTheory.Foundation.StructuralSentenceHierarchyV2.structuralV2_crown_family
Summits	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_no_fixed_structural_stratum	No fixed structural stratum + abstract org obstruction at height B.	NoveltyTheory.Ridge.OrganizationAbstractObstruction.organization_obstruction_supports_generalized_crown
Summits	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_future_defeat_of_terminality	FutureDefeat for natCounter.	NoveltyTheory.Models.StratifiedFinality.natCounter_futureDefeat
Summits	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_generator_truth_family	traceEq generator truths + structural ascent family.	NoveltyTheory.Foundation.StructuralSentenceHierarchyV2.structuralV2_crown_family trace lemmas
Summits	NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_class_transfer	Transfer of enum crown gap across admissible class.	NoveltyTheory.Ridge.CrownTransfer.crown_transfers_across_admissible_class

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Table 3 (continued)

Layer	Lean 4 declaration	Informal content	Immediate dependencies
Summits	`NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_not_model_artifact`	Non-single-model corollary.	`NoveltyTheory.Ridge.CrownTransfer.generalized_crown_not_single_model_artifact`
Ridge	`NoveltyTheory.Ridge.BroadTransfer.broad_transfer_of_generalized_crown`	Enum witness gap for every natural admissible instance (two observation kinds).	`NoveltyTheory.Foundation.PhaseGeneralizationFacts.carrier_outputEnum_gap_at` `NoveltyTheory.Core.NaturalAdmissibleClass.NaturalAdmissibleInstance`
Ridge	`NoveltyTheory.Ridge.BroadTransfer.crown_not_encoding_artifact_in_natural_class`	Bool-tag twins + parallel gaps in the natural class.	`NoveltyTheory.Ridge.BroadTransfer.multiple_nontrivially_distinct_instances`
Ridge	`NoveltyTheory.Ridge.UnboundedOrganization.natural_no_final_organization_theorem`	No final bounded natural organization certificate for full future.	`NoveltyTheory.Ridge.OrganizationHeightObstruction` `NoveltyTheory.Core.NaturalOrganization`
Ridge	`NoveltyTheory.Ridge.NaturalAntiClosure.natural_anti_closure_theorem`	Packaged anti-closure / disjunctive inevitability on natural numeric systems.	`NoveltyTheory.Ridge.NaturalAntiClosure.NaturalNumericSystem` weak naturality lemmas

D Dependency reference (Lean 4 modules and certificate chains)

D.1 Module import closure at the summit

`GeneralizedFinalCrownPackage` aggregates the following modules (direct imports), grouped by architectural role:

`Core primitives`

- `NoveltyTheory.Core.GenerativeSystem`
- `NoveltyTheory.Core.GeneralizedCarrier`
- `NoveltyTheory.Core.GeneratorTruth`
- `NoveltyTheory.Core.OrganizationAbstract`
- `NoveltyTheory.Core.Phase`
- `NoveltyTheory.Core.PhaseSyntaxGeneral`
- `NoveltyTheory.Core.SentenceSemantics`

`Foundation`

- `NoveltyTheory.Foundation.PhaseGeneralizationFacts`
- `NoveltyTheory.Foundation.StructuralSentenceHierarchyV2`

`Models`

- `NoveltyTheory.Models.ClassTransferExamples`
- `NoveltyTheory.Models.OutputEnumCrownFamily`
- `NoveltyTheory.Models.SignatureTower`
- `NoveltyTheory.Models.StructuralCrownFamily`
- `NoveltyTheory.Models.StratifiedFinality`

`Ridge`

- `NoveltyTheory.Ridge.CrownTransfer`
- `NoveltyTheory.Ridge.OrganizationAbstractObstruction`

`Summits (prior crown)`

- `NoveltyTheory.Summits.FinalCrownPackage`

Boundary certificates (minimality / numeric trace). These modules are not in the `GeneralizedFinalCrownPackage` import closure but supply sharp trust-boundary lemmas cited in the main text: `NoveltyTheory.Core.MinimalHypotheses`, `NoveltyTheory.Ridge.ConservativeSeparationCountermodel`, `NoveltyTheory.Ridge.ClosureCollapseBoundary`, `NoveltyTheory.Ridge.UniversalUpwardNecessity`.

Imports adjacent to `GeneralizedFinalCrownPackage` (natural class extension). The following modules supply broad transfer and organization / anti-closure certificates alongside the six crown clauses:

`Core`

- `NoveltyTheory.Core.NaturalityAxioms`
- `NoveltyTheory.Core.StructuralSentenceNaturality`
- `NoveltyTheory.Core.NaturalAdmissibleClass`
- `NoveltyTheory.Core.NaturalOrganization`

`Foundation`

- `NoveltyTheory.Foundation.NaturalityFacts`
- `NoveltyTheory.Foundation.StructuralSentenceNaturalityFacts`

`Models`

- `NoveltyTheory.Models.NaturalClassExamples`

`Ridge`

- `NoveltyTheory.Ridge.AdmissibilityNaturality`
- `NoveltyTheory.Ridge.BroadTransfer`
- `NoveltyTheory.Ridge.UnboundedOrganization`
- `NoveltyTheory.Ridge.NaturalAntiClosure`

Summits

- NoveltyTheory.Summits.BroadTransferSummit
- NoveltyTheory.Summits.UnboundedOrganizationSummit
- NoveltyTheory.Summits.NaturalAntiClosureSummit

Transitive imports pull in Mathlib and, among project modules, the following (illustrative, not exhaustive):

- NoveltyTheory.Ridge.OrganizationHeightObstruction
- NoveltyTheory.Ridge.RetroStructuralGap
- NoveltyTheory.Models.InvariantTower
- NoveltyTheory.Foundation.RetroStructuralTruthV2
- NoveltyTheory.Models.SentenceProvability

The complete `import` graph is machine-readable; this list is a concise index.

D.2 Certificate chains (summit → ingredients)

Each named summit theorem below unfolds to the listed Lean 4 certificates; indented arrows mark parallel immediate dependencies.

- **Iterated structural ascent.**

  NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_iterated_structural_ascent
  ↪ NoveltyTheory.Foundation.StructuralSentenceHierarchyV2.structuralV2_crown_family.

- **No fixed structural stratum.**

  NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_no_fixed_structural_stratum
  ↪ NoveltyTheory.Ridge.OrganizationAbstractObstruction.organization_obstruction_supports_generalized_crown
  ↪ NoveltyTheory.Models.SignatureTower.finite_signature_cannot_organize_full_ladder
  ↪ NoveltyTheory.Ridge.OrganizationHeightObstruction.no_finite_adequate_organization_totalizes_future.

- **Future defeat of terminality.**

  NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_future_defeat_of_terminality
  ↪ NoveltyTheory.Models.StratifiedFinality.natCounter_futureDefeat.

- **Generator-truth family.**

  NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_generator_truth_family
  ↪ NoveltyTheory.Foundation.StructuralSentenceHierarchyV2.structuralV2_crown_family *(structural ascent family)*
  ↪ generator-structural `traceEq` facts (`isGeneratorStructural_traceEq`, `holdsAt_natCounter_traceEq_self`).

- **Class transfer.**

  NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_class_transfer
  ↪ NoveltyTheory.Ridge.CrownTransfer.crown_transfers_across_admissible_class
  ↪ NoveltyTheory.Ridge.CrownTransfer.true_general_phase_gap_at /

alt_general_phase_gap_at
↪ NoveltyTheory.Models.OutputEnumCrownFamily.outputEnumCrownWitness_proves_succ_not_at
↪ *carrier embeddings* (phase embed / faithful pullback).

- **Not a single-model artifact.**

  NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_not_model_artifact
  ↪ NoveltyTheory.Ridge.CrownTransfer.generalized_crown_not_single_model_artifact
  Constructive data: trueBoolProdNat, altBoolProdNat, alignment lemmas, first-time trace mismatch.

- **Broad transfer across natural admissible instances.**

  NoveltyTheory.Ridge.BroadTransfer.broad_transfer_of_generalized_crown
  ↪ NoveltyTheory.Ridge.BroadTransfer.naturalClass_preserves_structural_ascent
  ↪ NoveltyTheory.Foundation.PhaseGeneralizationFacts.carrier_outputEnum_gap_at
  ↪ NoveltyTheory.Core.NaturalAdmissibleClass.NaturalAdmissibleInstance.

- **Non-encoding artifact in the natural class.**

  NoveltyTheory.Ridge.BroadTransfer.crown_not_encoding_artifact_in_natural_class
  ↪ NoveltyTheory.Ridge.BroadTransfer.multiple_nontrivially_distinct_instances.

- **Bounded natural organization obstruction.**

  NoveltyTheory.Ridge.UnboundedOrganization.natural_no_final_organization_theorem
  ↪ NoveltyTheory.Ridge.UnboundedOrganization.weakNaturalOrganization_bounded_obstruction
  ↪ NoveltyTheory.Ridge.OrganizationAbstractObstruction.no_bounded_abstractOrganization_totalizes_future.

- **Natural anti-closure packaging.**

  NoveltyTheory.Ridge.NaturalAntiClosure.natural_anti_closure_theorem
  ↪ NoveltyTheory.Ridge.NaturalAntiClosure.weak_natural_conditions_imply_nonfinality_or_ascent.

D.3 Conceptual dependency layers (DAG summary)

Height	Role
0	**Ontology**: generative systems, phases, sentences, carrier alignment.
1	**Dynamics**: natCounter, trace identities, singleton phases, regimes regimeUpto.
2	**Logic at depth**: ProvesAt / InvariantTower, diagonal obstruction not_proves_trace_diag.
3	**Witness families**: outputEnumCrownWitness, histSeqUpto strict ascent.
4	**Obstructions**: finite signature vs. full ladder; finite organization height vs. total future.
5	**Transfer**: Bool×ℕ twins, faithful phase embed, ProvesAtCarrier pullback.
6	**Summit**: GeneralizedFinalCrownPackage re-exports.
7	**Strengthening**: BroadTransfer, UnboundedOrganization, NaturalAntiClosure + summits.
8	**Minimality / numeric boundaries**: packaged paradigm shift vs. weak step + ConservativeOver; FutureDefeat vs. uniform bounded numeric trace; explicit counterexamples to unrestricted universal glosses.

Heights are expository only: many formal edges cross layers (for example semantic lemmas for HoldsAt).

E Obstructions and blocked universals

Explicit obstruction lemmas refute naive universal diagonal schemes, naive "infinite paradigm tower for every regime family" glosses, and the idea that lawful numeric generators must universally realize unbounded observable recurrence. Together with conservativity–separation witnesses and the numeric `FutureDefeat` equivalence to uniform boundedness (Table 3), these results fix the trust boundary of the admissible predicates. The natural-class theorems in Table 2 are *not* blanket universals: each statement carries explicit hypotheses (`NaturalAdmissibleInstance`, `NaturalBoundedOrganization`, `NaturalNumericSystem`, weak naturality). Any informal "inevitability" claim must be read against that quantifier structure.

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